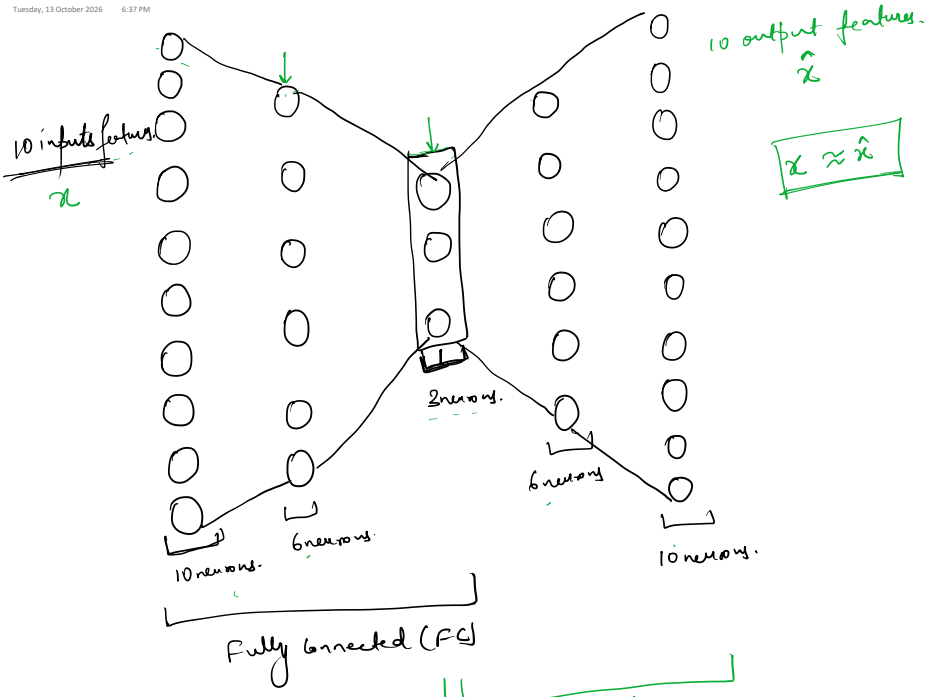
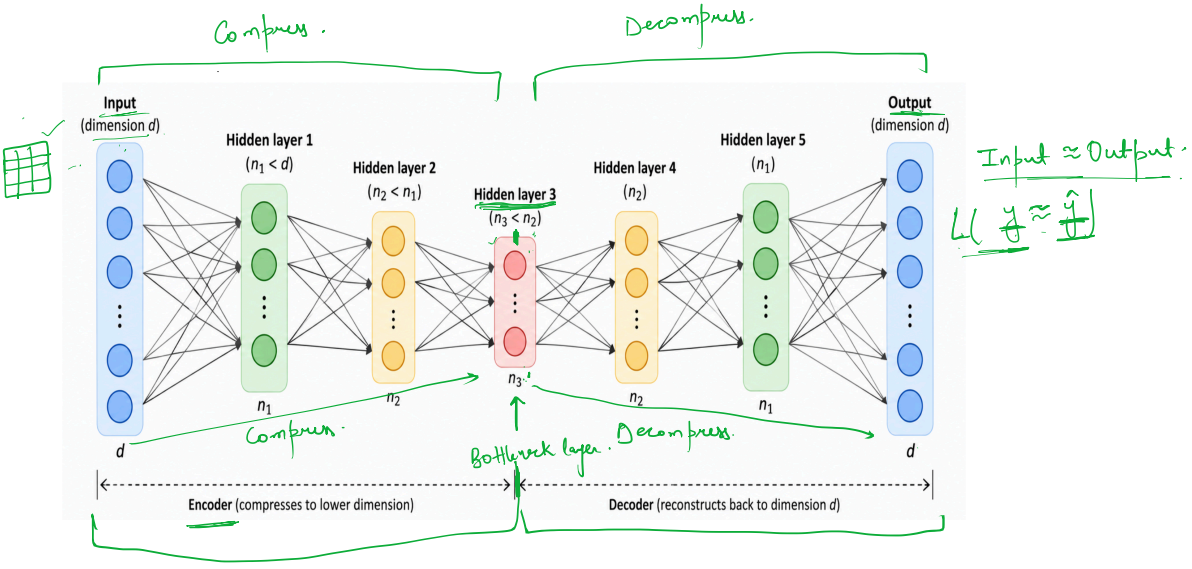
Autoencoders: Learning to Compress and Reconstruct

Tuesday, 13 October 2026 6:37 PM



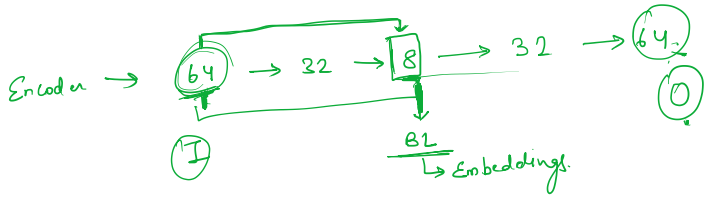


Autoencoder → Encoder + Bottleneck layer + Decoder.

↓

Compressed Representation.

Embedding.



L (I, O)

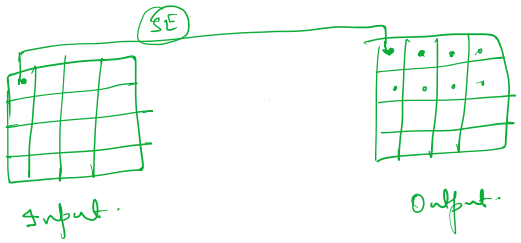
↓

Reduce.

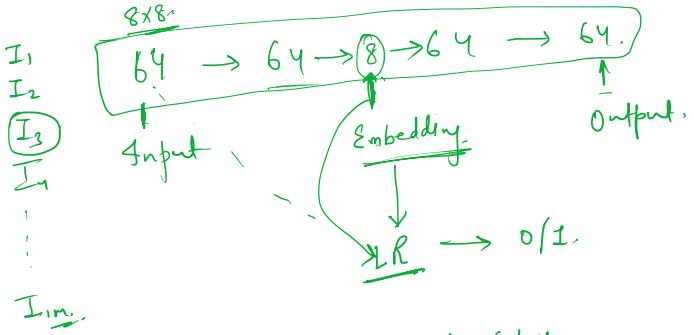
PCA → Principal Component Analysis.

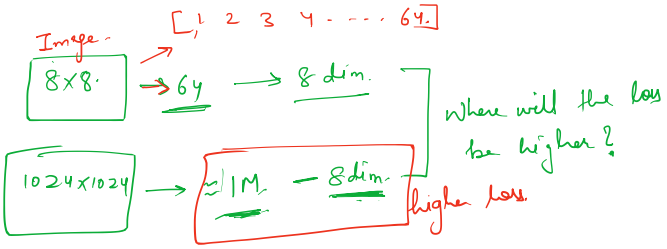
n_component. → You extract n directions with maximum variance.

100 → 5



MSE for the whole image.



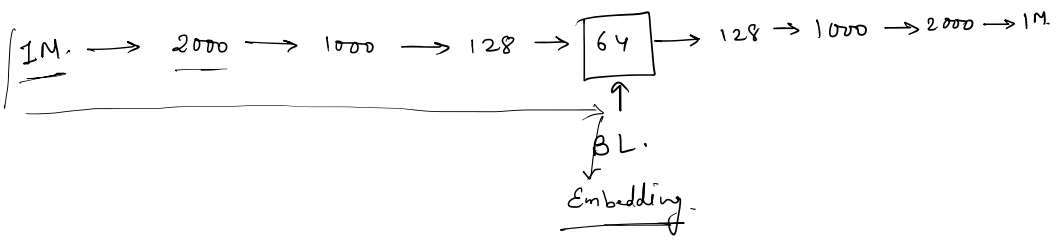


Recommender Systems.

I3 E → 8 dim.

Cosine similarity (I3E, I10E) → User.

* Transfer Learning.

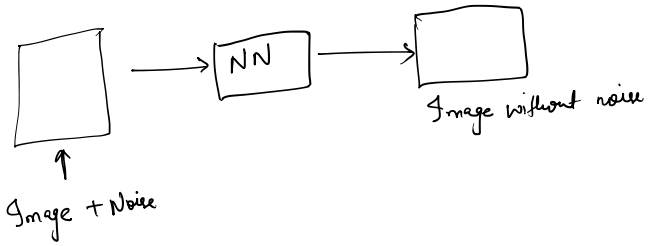


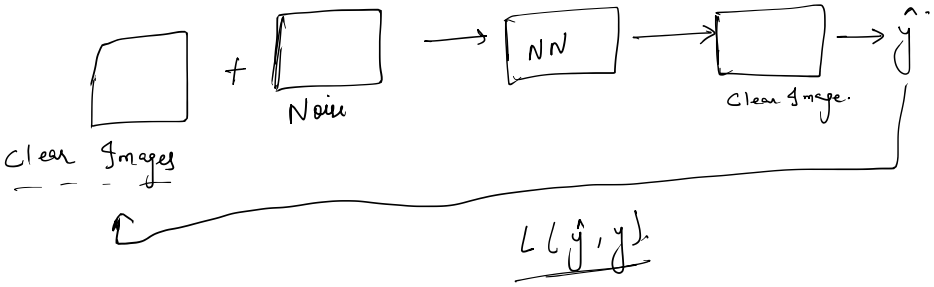
Reusing a network trained for one task as a feature extractor for a different, related task is called **Transfer Learning**.

* Denoising

Try to remove the noise from an image.

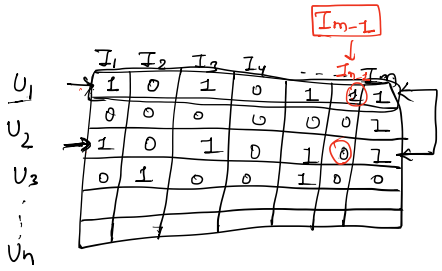
U





* Recommendation System Using Autoencoder.

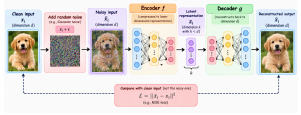
Netflix.



U1 & U3 are similar.

Netflix → Millions of Users. → Sparse.
→ 100000's of movies.

① Raw Vectors → Very sparse.
② Reduced Dim. Vectors. ✓ → Autoencoders


Clean input
x_i
(dimension d)
Add random noise
(e.g., Gaussian noise)
$x_i + \epsilon$
Noisy input
$\tilde{x}_i$
(dimension d)
Encoder f
Compresses to lower-dimensional representation
Latent representation
h_i
(dimension f with $f < d$)
Decoder g
Expands back to its original dimension d
Reconstructed output
$\hat{x}_i$
(dimension d)
Compare with clean input (not the noisy one)
$\mathcal{L} = \|\hat{x}_i - x_i\|^2$
e.g., MSE loss